**CMPT353 Project Test Report**
Matt Berry
jxa417
11356787

-

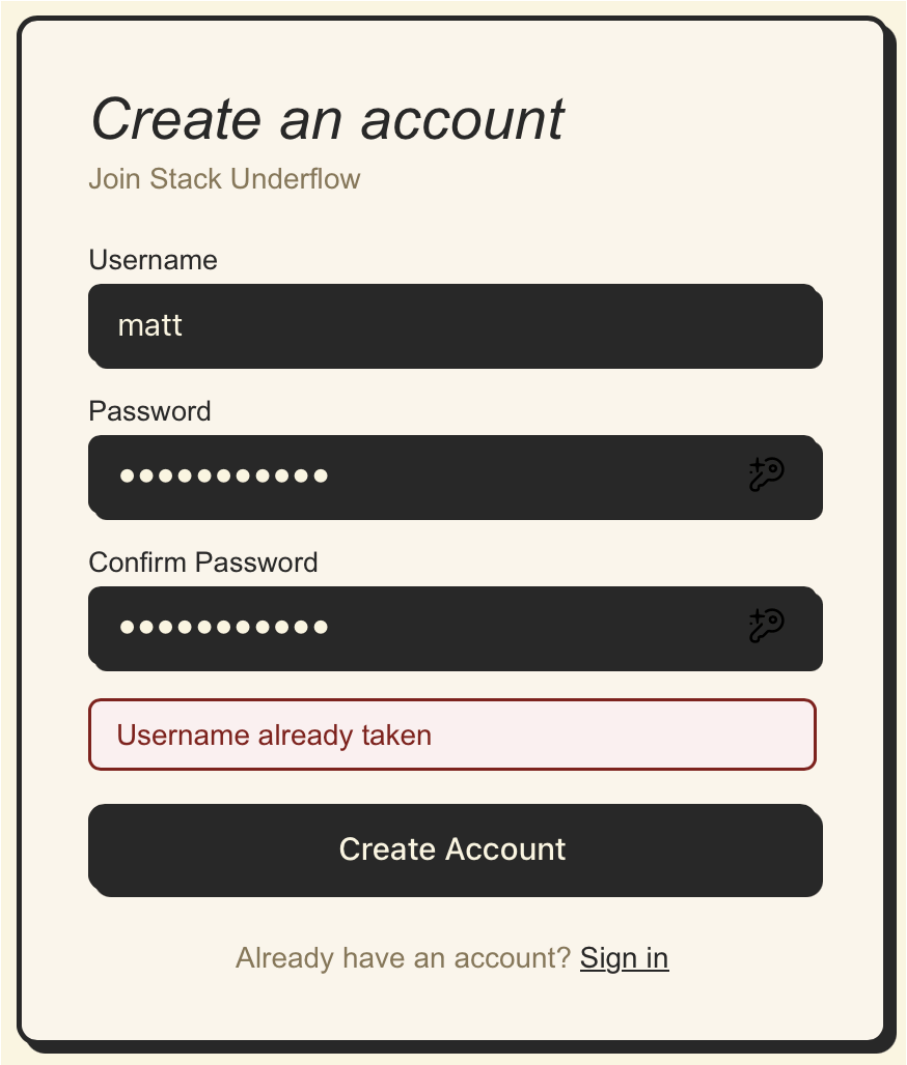
The following test cases include the expected result, steps performed, actual results, docker outputs and any necessary inputs when applicable. These tests were performed manually to verify that the system conforms to the project specifications and to attempt to expose any poorly handled errors or gaps in the UX.

-

**TC-01: Sign up with valid data**
**Expected:** user created, redirected, username shown in top banner
**Steps:** started the program, navigated to sign up, entered fields and pressed create account.
**Inputs:** username: matt, password: password123
**Actual:** user created successfully, redirected to /channels, username shown in top banner
**Output:**

```
app-1    | GET /createaccount 200 in 329ms (compile: 319ms, render: 11ms)
app-1    | POST /api/auth/signup 201 in 373ms (compile: 152ms, render: 221ms)
app-1    | GET /api/auth/providers 200 in 618ms (compile: 612ms, render: 6ms)
app-1    | GET /api/auth/csrf 200 in 9ms (compile: 5ms, render: 3ms)
app-1    | POST /api/auth/callback/credentials? 200 in 183ms (compile: 1527µs, render: 181ms)
app-1    | GET /api/auth/session 200 in 6ms (compile: 1282µs, render: 5ms)
app-1    | GET /channels 200 in 289ms (compile: 281ms, render: 8ms)
app-1    | GET /channels 200 in 29ms (compile: 1428µs, render: 28ms)
app-1    | GET /api/channels 200 in 96ms (compile: 86ms, render: 10ms)
app-1    | GET /api/channels 200 in 98ms (compile: 90ms, render: 9ms)
```

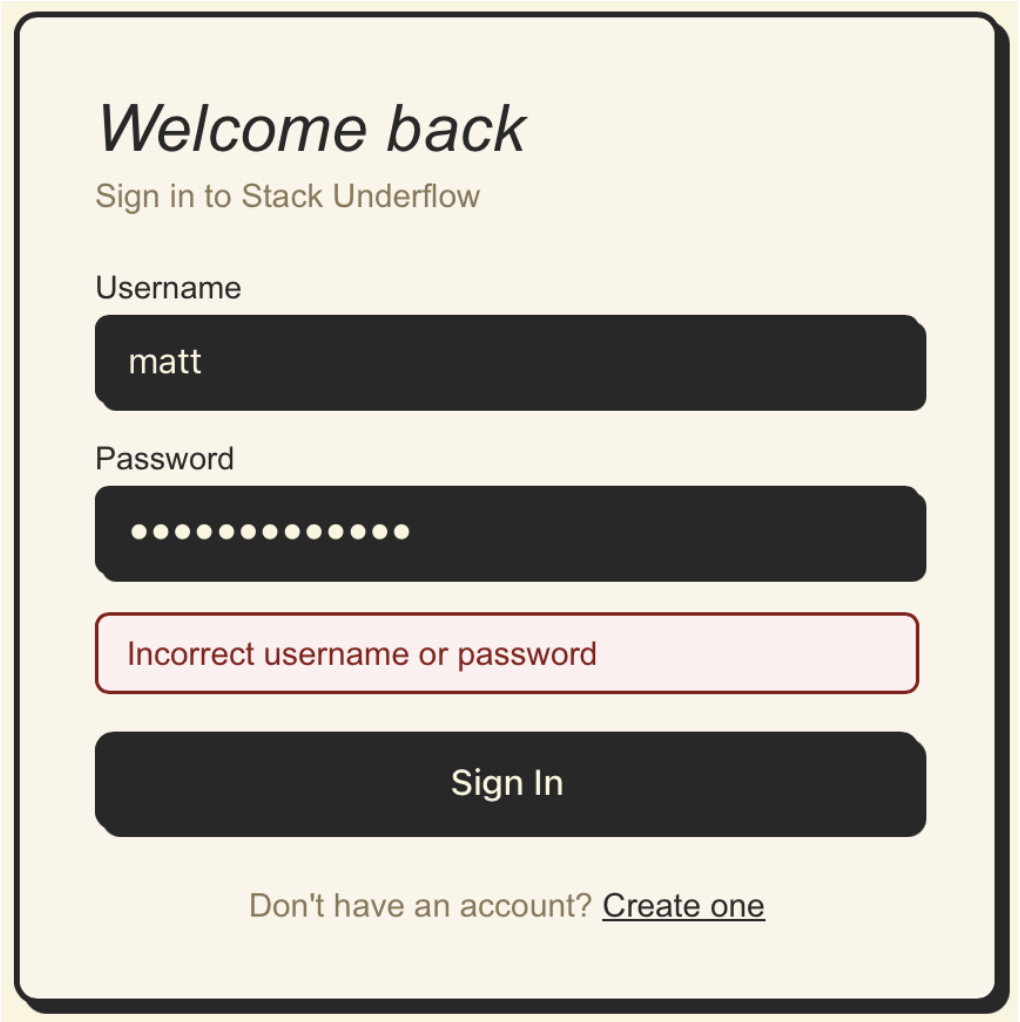
**TC-02: Sign up with duplicate username**
**Expected:** clear error shown
**Steps:** Logged out of previous session, attempted to create account with same username
**Actual:** clear error shown




**Output:**

```
app-1   | GET /api/auth/csrf 200 in 57ms (compile: 46ms, render: 11ms)
app-1   | POST /api/auth/signout 200 in 8ms (compile: 1852µs, render: 7ms)
app-1   | GET / 200 in 58ms (compile: 5ms, render: 53ms)
app-1   | GET /createaccount 200 in 32ms (compile: 6ms, render: 26ms)
app-1   | POST /api/auth/signup 409 in 28ms (compile: 4ms, render: 24ms)
```

**TC-03: Sign in with wrong password**
**Expected:** access denied, error shown
**Steps:** navigated to log in, entered credentials and pressed log in
**Inputs:** username: matt, password: notmypassword
**Actual:** error shown, access denied




**Output:**

```
app-1    | GET /login 200 in 442ms (compile: 425ms, render: 17ms)
app-1    | GET /api/auth/providers 200 in 49ms (compile: 35ms, render: 14ms)
app-1    | GET /api/auth/csrf 200 in 7ms (compile: 2ms, render: 5ms)
app-1    | [auth][error] CredentialsSignin: Read more at
https://errors.authjs.dev#credentialssignin
app-1    |     at Module.callback
(/app/.next/dev/server/chunks/node_modules_@auth_core_36df0127._.js:4314:30)
app-1    |     at async AuthInternal
(/app/.next/dev/server/chunks/node_modules_@auth_core_36df0127._.js:5148:24)
app-1    |     at async Auth
(/app/.next/dev/server/chunks/node_modules_@auth_core_36df0127._.js:5370:34)
app-1    |     at async AppRouteRouteModule.do
(/app/node_modules/next/dist/compiled/next-server/app-route-turbo.runtime.dev.js:5:
37866)
app-1    |     at async AppRouteRouteModule.handle
(/app/node_modules/next/dist/compiled/next-server/app-route-turbo.runtime.dev.js:5:
45156)
app-1    |     at async responseGenerator
(/app/.next/dev/server/chunks/node_modules_next_c0d7f167._.js:17154:38)
```

app-1 | at async AppRouteRouteModule.handleResponse (/app/node_modules/next/dist/compiled/next-server/app-route-turbo.runtime.dev.js:1:191938)
app-1 | at async handleResponse (/app/.next/dev/server/chunks/node_modules_next_c0d7f167._.js:17217:32)
app-1 | at async Module.handler (/app/.next/dev/server/chunks/node_modules_next_c0d7f167._.js:17270:13)
app-1 | at async DevServer.renderToResponseWithComponentsImpl (/app/node_modules/next/dist/server/base-server.js:1442:9)
app-1 | at async DevServer.renderPageComponent (/app/node_modules/next/dist/server/base-server.js:1494:24)
app-1 | at async DevServer.renderToResponseImpl (/app/node_modules/next/dist/server/base-server.js:1544:32)
app-1 | at async DevServer.pipeImpl (/app/node_modules/next/dist/server/base-server.js:1038:25)
app-1 | at async NextNodeServer.handleCatchallRenderRequest (/app/node_modules/next/dist/server/next-server.js:395:17)
app-1 | at async DevServer.handleRequestImpl (/app/node_modules/next/dist/server/base-server.js:929:17)
app-1 | at async /app/node_modules/next/dist/server/dev/next-dev-server.js:387:20
app-1 | at async Span.traceAsyncFn (/app/node_modules/next/dist/trace/trace.js:157:20)
app-1 | at async DevServer.handleRequest (/app/node_modules/next/dist/server/dev/next-dev-server.js:383:24)
app-1 | at async invokeRender (/app/node_modules/next/dist/server/lib/router-server.js:248:21)
app-1 | at async handleRequest (/app/node_modules/next/dist/server/lib/router-server.js:447:24)
app-1 | at async requestHandlerImpl (/app/node_modules/next/dist/server/lib/router-server.js:496:13)
app-1 | at async Server.requestListener (/app/node_modules/next/dist/server/lib/start-server.js:226:13)
app-1 | POST /api/auth/callback/credentials? 200 in 244ms (compile: 1920µs, render: 242ms)
app-1 | GET /api/auth/session 200 in 4ms (compile: 1342µs, render: 3ms)

**TC-04: Create channel while logged out**

**Expected:** 401 returned, UI blocked

**Steps:** navigated to channels list without logging in, create channels button blocked

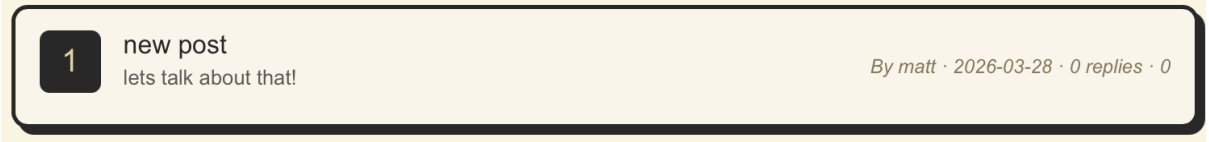
**Actual:** create channels button does not render, no way to manipulate URL to create a channel without credentials.

**Output:**

app-1 | GET /channels 200 in 49ms (compile: 25ms, render: 24ms)
app-1 | GET /api/channels 200 in 35ms (compile: 3ms, render: 32ms)
app-1 | GET /api/channels 200 in 28ms (compile: 1943µs, render: 26ms)

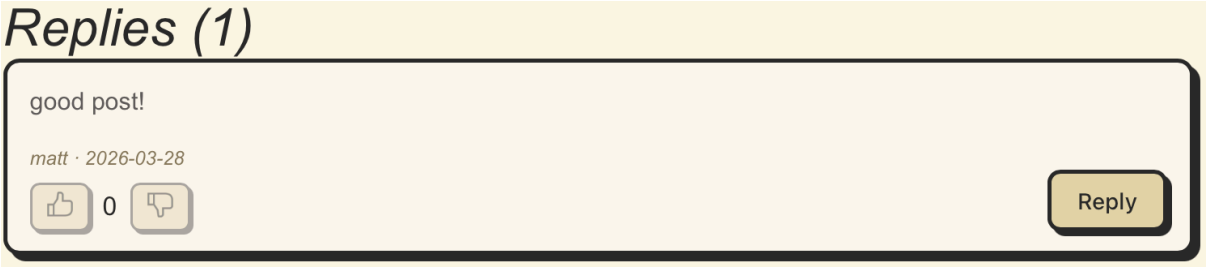
**TC-05: Create channel while logged in**
**Expected:** appears in channels list
**Inputs:** name: code, description: coding and whatnot
**Actual:** channel created, appears in list
**Output:**
app-1 | GET /channels 200 in 43ms (compile: 16ms, render: 28ms)
app-1 | GET /api/channels 200 in 52ms (compile: 3ms, render: 49ms)
app-1 | GET /api/channels 200 in 58ms (compile: 24ms, render: 34ms)
app-1 | POST /api/channels 201 in 51ms (compile: 17ms, render: 35ms)

**TC-06: Create post in channel while logged in**
**Expected:** appears with author name at current time
**Steps:** navigated into channel, new post, filled fields and pressed create
**Input:** title: new post, body: lets talk about that!
**Actual:** post created, appears in channel



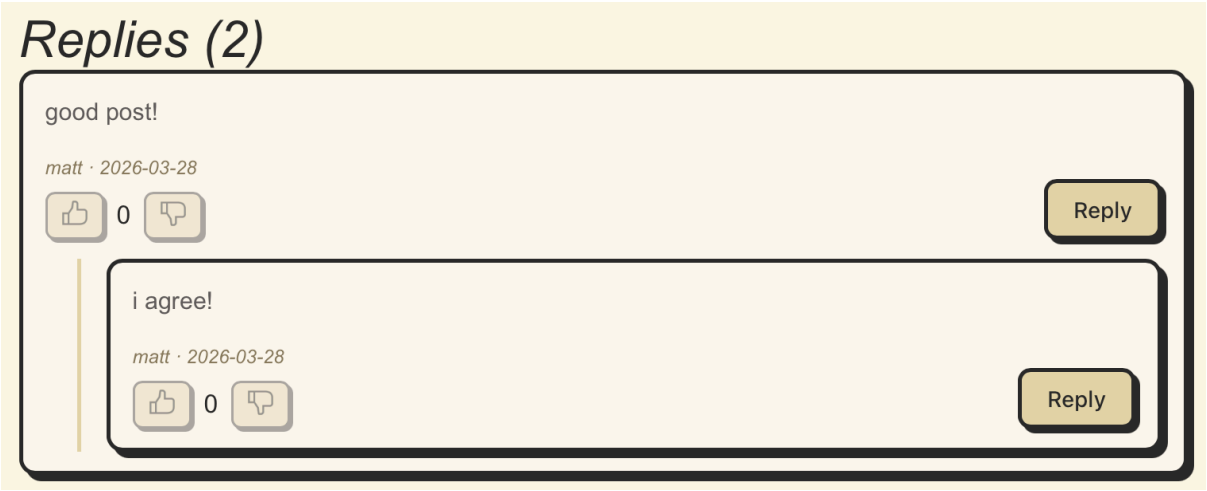

**Output**:
app-1 | GET /channels/code 200 in 53ms (compile: 35ms, render: 18ms)
app-1 | GET /api/channels/code 200 in 68ms (compile: 28ms, render: 40ms)
app-1 | GET /api/channels/code 200 in 66ms (compile: 41ms, render: 25ms)
app-1 | POST /api/channels/code 201 in 42ms (compile: 24ms, render: 18ms)

**TC-07: Reply to a post while logged in**
**Expected:** reply shown threaded
**Steps:** navigate into post, press reply, fill out field and press submit reply
**Input:** great post!
**Actual:** reply created under main post



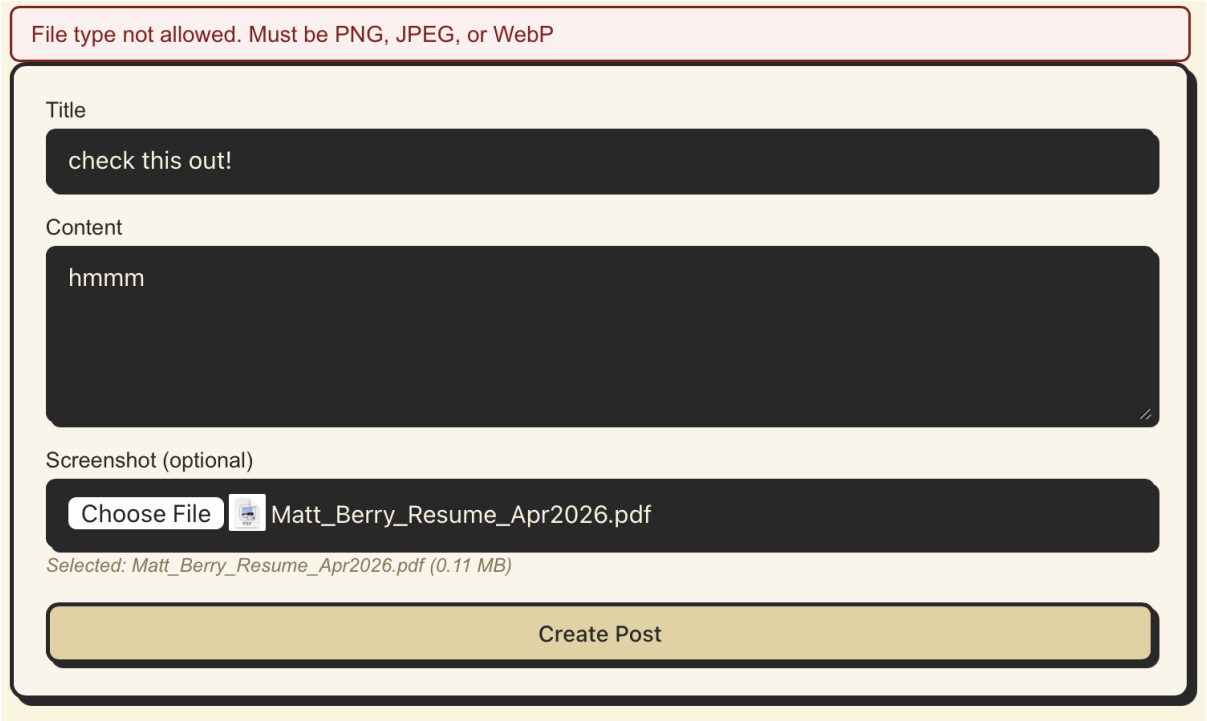

**Output:**
app-1 | GET /channels/code/posts/7 200 in 51ms (compile: 32ms, render: 19ms)
app-1 | GET /api/channels/code/posts/7 200 in 76ms (compile: 41ms, render: 35ms)
app-1 | GET /api/channels/code/posts/7 200 in 25ms (compile: 4ms, render: 21ms)
app-1 | POST /api/channels/code/posts/7 201 in 30ms (compile: 12ms, render: 18ms)

**TC-08: Reply to a reply**
**Expected:** nested correctly
**Steps:** replied to previous reply, filled field and pressed submit reply
**Input:** i agree!
**Actual:** reply created with proper nesting




**Output:**
app-1 | POST /api/channels/code/posts/7 201 in 28ms (compile: 10ms, render: 17ms)

**TC-09: Upload an invalid file type**
**Expected:** rejected with error
**Steps:** navigated to channel ‘code’ as previously created, pressed new post, filled fields and uploaded a pdf
**Inputs:** title: check this out!, body: hmmmm, attachment: Matt_Berry_Resume_Apr2026.pdf
**Actual:** clear error displayed, post creation rejected

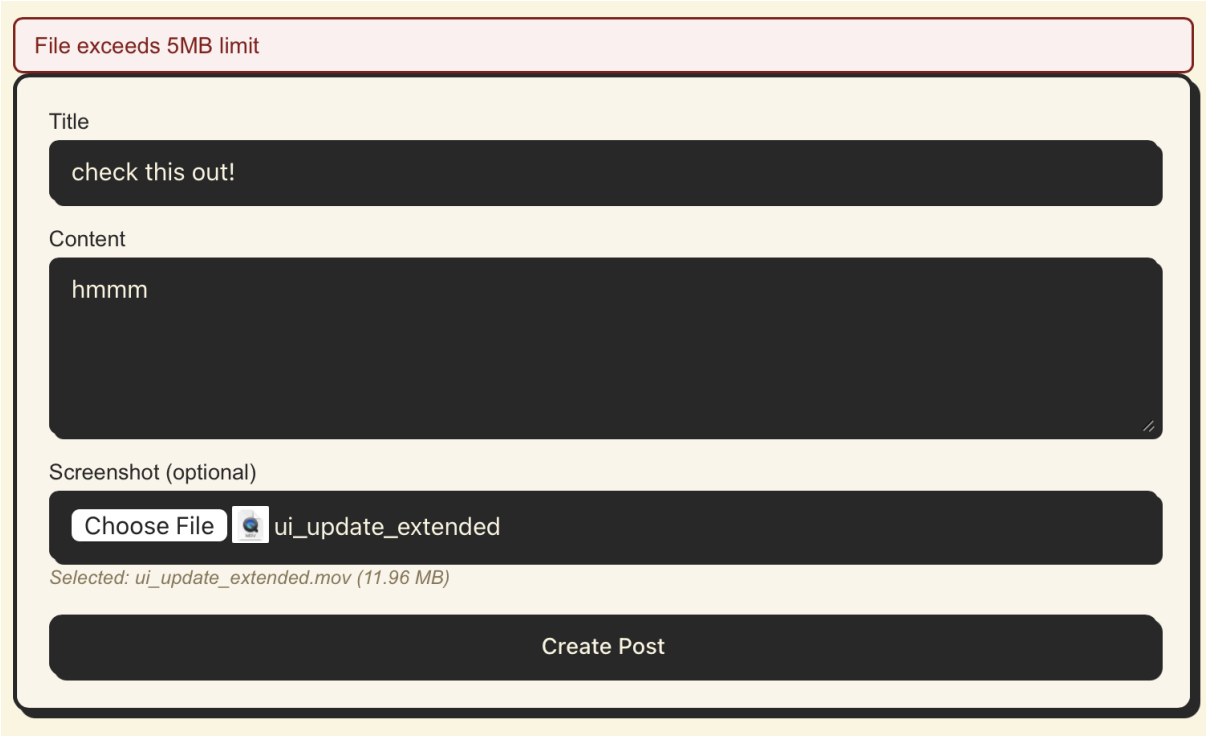



**Output:**

```
app-1    |  GET /api/channels/code 200 in 8ms (compile: 3ms, render: 6ms)
app-1    |  POST /api/channels/code 400 in 34ms (compile: 24ms, render: 10ms)
```

**TC-10: Upload image > 5 MB**
**Expected:** rejected with error
**Steps:** swapped previous upload with a large file
**Input:** title: check this out!, body: hmmm, attachment: ui_update_extended

**Actual:** post creation rejected, error shown. Here the file type is also invalid, but the server checks the size first before the type, allowing us to check this behaviour without needing a massive .png file.




**Output:**
app-1 | POST /api/channels/code 400 in 151ms (compile: 38ms, render: 113ms)

**TC-11: Upload valid PNG**
**Expected:** displays inline with post
**Steps:** swapped attachment with valid png, created post, navigated into post
**Inputs:** title: check this out!, body: hmmm, attachment: fitpic.png

**Actual:** Image shows as intended (from Google images)

**Output:**

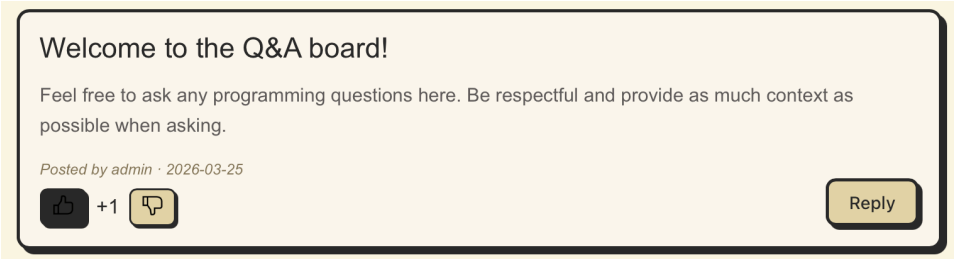
app-1 | POST /api/channels/code 201 in 98ms (compile: 31ms, render: 66ms)
app-1 | GET /channels/code/posts/8 200 in 41ms (compile: 24ms, render: 17ms)
app-1 | GET /api/channels/code/posts/8 200 in 24ms (compile: 12ms, render: 12ms)
app-1 | GET /api/channels/code/posts/8 200 in 26ms (compile: 9ms, render: 16ms)
app-1 | GET /api/attachments/2 200 in 17ms (compile: 9ms, render: 7ms)

**TC-12: Upvote a post**

**Expected:** score increases by 1

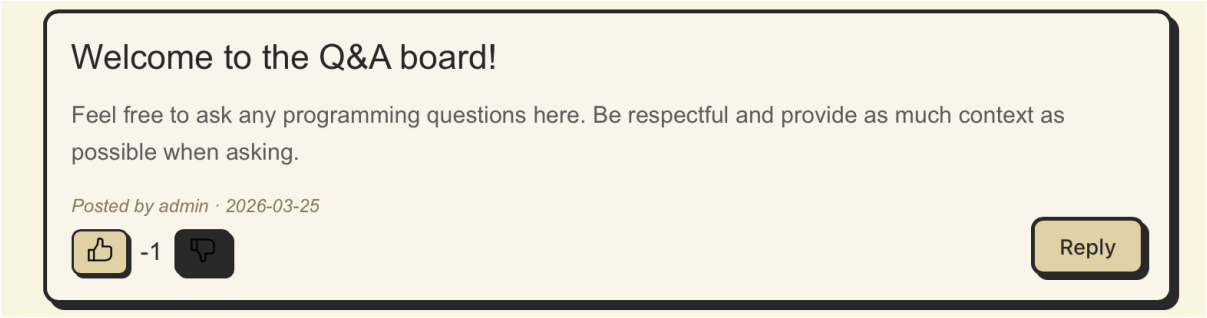
**Steps:** navigated to channels > general > welcome (from database seed), upvoted post

**Actual:** button press updated vote counter, shaded appearance to suggest pressed



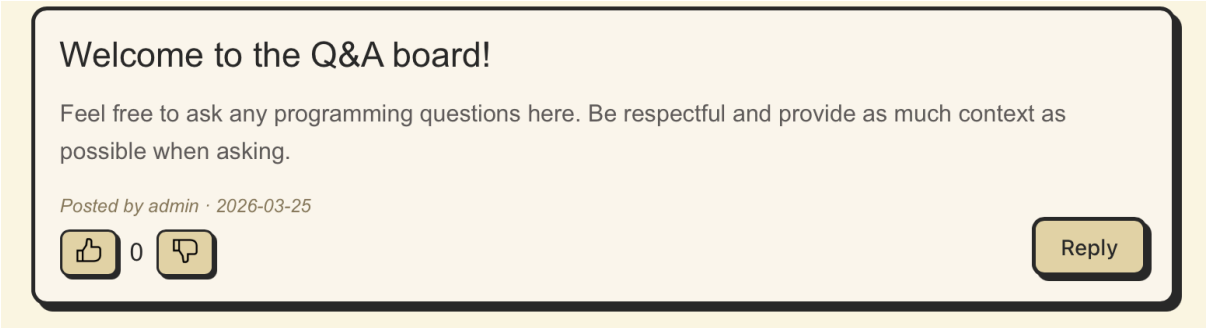
**Output:**
app-1 | GET / 200 in 66ms (compile: 28ms, render: 39ms)
app-1 | GET /channels 200 in 34ms (compile: 8ms, render: 26ms)
app-1 | GET /api/channels 200 in 78ms (compile: 13ms, render: 66ms)
app-1 | GET /api/channels 200 in 27ms (compile: 6ms, render: 21ms)
app-1 | GET /channels/general 200 in 48ms (compile: 33ms, render: 15ms)
app-1 | GET /api/channels/general 200 in 21ms (compile: 14ms, render: 6ms)
app-1 | GET /api/channels/general 200 in 22ms (compile: 11ms, render: 11ms)
app-1 | GET /channels/general/posts/1 200 in 38ms (compile: 21ms, render: 17ms)
app-1 | GET /api/channels/general/posts/1 200 in 25ms (compile: 9ms, render: 16ms)
app-1 | GET /api/channels/general/posts/1 200 in 26ms (compile: 12ms, render: 14ms)
app-1 | POST /api/votes 200 in 168ms (compile: 158ms, render: 10ms)

**TC-13: Change vote from up to down**
**Expected:** score decreases by 2
**Steps:** changed vote from up to down on same post
**Actual:** score altered from +1 to -1



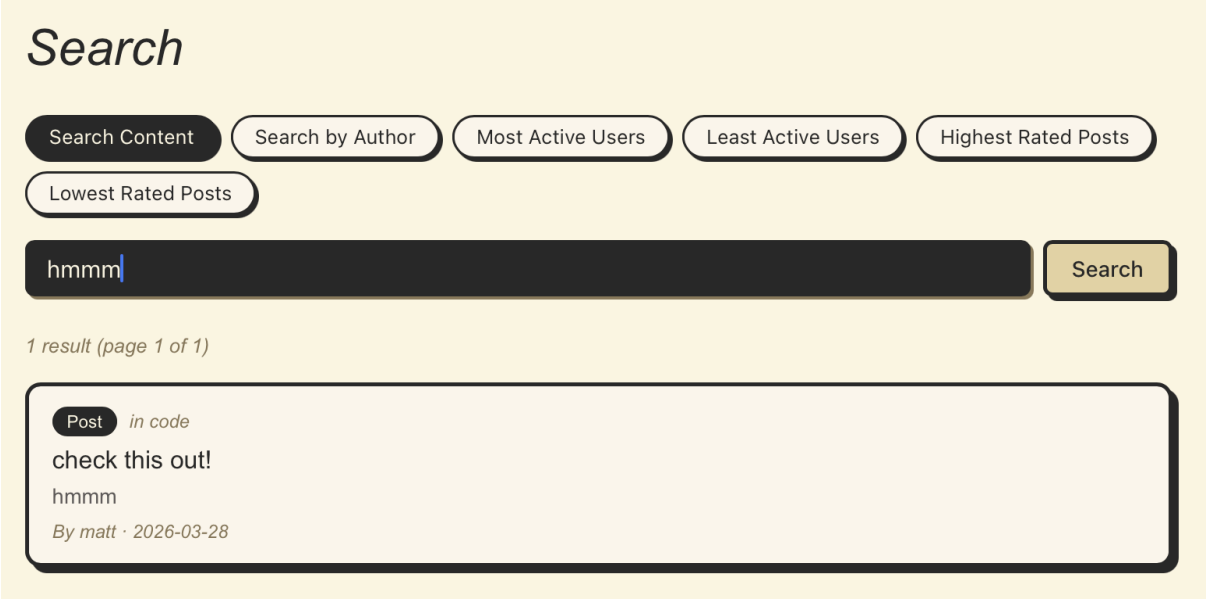

**Output:**
app-1 | POST /api/votes 200 in 90ms (compile: 34ms, render: 56ms)

**TC-14: Remove vote**
**Expected:** score returns to original
**Steps:** pressed downvote button again
**Actual:** score returns to original, neither button shows it was pressed



**Output:**
app-1 | POST /api/votes 200 in 40ms (compile: 6ms, render: 34ms)

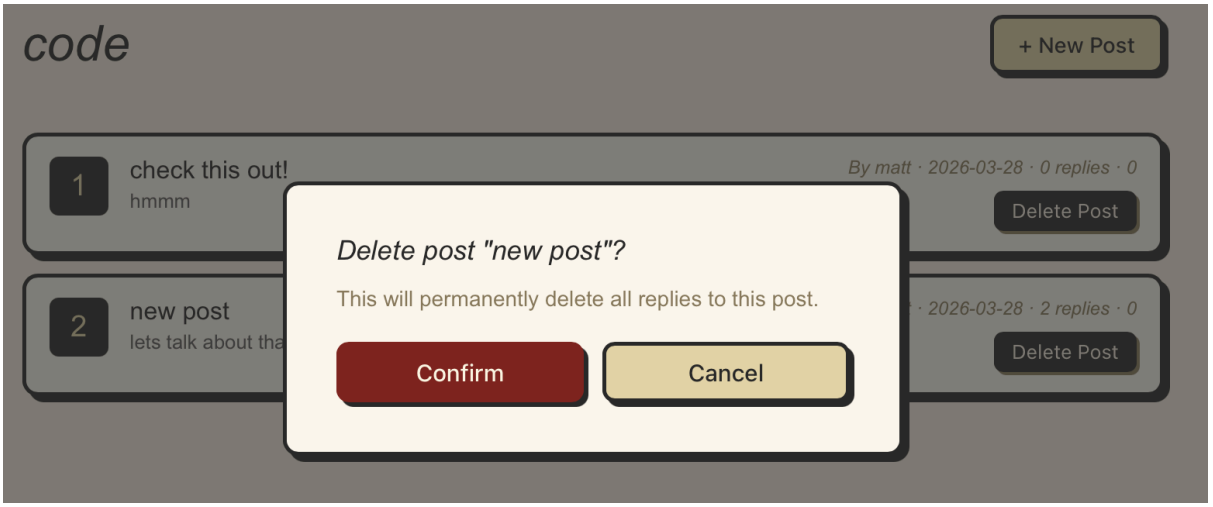
**TC-15: Search for keyword present in a post**
**Expected:** result appears with pressable link, link navigates to post
**Steps:** entered input in search bar and pressed search, pressed link of first post in results list
**Input:** hmmm
**Actual:** filling the search field and pressing search navigates to the search page with results shown. Results are clickable links that navigate to the post found.




**Output:**
app-1 | GET /search?q=hmmm 200 in 45ms (compile: 20ms, render: 25ms)
app-1 | GET /api/search?q=hmmm&type=content&page=1 200 in 34ms (compile: 3ms, render: 32ms)
app-1 | GET /api/search?q=hmmm&type=content&page=1 200 in 29ms (compile: 8ms, render: 21ms)
app-1 | GET /channels/code/posts/8 200 in 50ms (compile: 33ms, render: 17ms)
app-1 | GET /api/channels/code/posts/8 200 in 22ms (compile: 1347µs, render: 21ms)
app-1 | GET /api/channels/code/posts/8 200 in 58ms (compile: 13ms, render: 45ms)
app-1 | GET /api/attachments/2 200 in 31ms (compile: 10ms, render: 22ms)

**TC-16: Admin deletes a post**
**Expected:** post removed for all users, verification pop up appears on delete
**Steps:** from home page, login as admin with seeded credentials, navigate to code channel, delete post ‘new post’. Logout of admin, login in as matt, navigate to code channel, verify post is deleted.

**Actual:** admin login success, delete post displays popup to verify action, logout successful, login as user successful, deleted post no longer exists in channel, total posts in channel updated accordingly.




**Output:**

```
app-1    | GET /api/auth/csrf 200 in 49ms (compile: 37ms, render: 12ms)
app-1    | POST /api/auth/signout 200 in 9ms (compile: 2ms, render: 7ms)
app-1    | GET / 200 in 98ms (compile: 6ms, render: 92ms)
app-1    | GET /login 200 in 41ms (compile: 18ms, render: 24ms)
app-1    | GET /api/auth/providers 200 in 22ms (compile: 18ms, render: 4ms)
app-1    | GET /api/auth/csrf 200 in 5ms (compile: 1741µs, render: 4ms)
app-1    | POST /api/auth/callback/credentials? 200 in 209ms (compile: 1877µs,
render: 208ms)
app-1    | GET /api/auth/session 200 in 5ms (compile: 1084µs, render: 4ms)
app-1    | GET /channels 200 in 9ms (compile: 1540µs, render: 8ms)
app-1    | GET /channels 200 in 29ms (compile: 6ms, render: 24ms)
app-1    | GET /api/channels 200 in 9ms (compile: 1310µs, render: 7ms)
app-1    | GET /api/channels 200 in 23ms (compile: 6ms, render: 17ms)
app-1    | GET /channels/code 200 in 40ms (compile: 25ms, render: 15ms)
app-1    | GET /api/channels/code 200 in 44ms (compile: 16ms, render: 28ms)
app-1    | GET /api/channels/code 200 in 47ms (compile: 26ms, render: 21ms)
app-1    | DELETE /api/admin/posts/7 200 in 566ms (compile: 552ms, render:
14ms)
app-1    | GET /api/auth/csrf 200 in 41ms (compile: 31ms, render: 9ms)
app-1    | POST /api/auth/signout 200 in 8ms (compile: 2ms, render: 6ms)
app-1    | GET / 200 in 39ms (compile: 2ms, render: 37ms)
app-1    | GET /login 200 in 25ms (compile: 5ms, render: 20ms)
app-1    | GET /api/auth/providers 200 in 30ms (compile: 24ms, render: 6ms)
app-1    | GET /api/auth/csrf 200 in 5ms (compile: 1612µs, render: 3ms)
app-1    | POST /api/auth/callback/credentials? 200 in 190ms (compile: 1574µs,
render: 188ms)
app-1    | GET /api/auth/session 200 in 5ms (compile: 1095µs, render: 4ms)
app-1    | GET /channels 200 in 8ms (compile: 1451µs, render: 7ms)
```

```
app-1    |  GET /channels 200 in 35ms (compile: 1396µs, render: 33ms)
app-1    |  GET /api/channels 200 in 6ms (compile: 1430µs, render: 5ms)
app-1    |  GET /api/channels 200 in 13ms (compile: 3ms, render: 10ms)
app-1    |  GET /channels/code 200 in 41ms (compile: 26ms, render: 15ms)
app-1    |  GET /api/channels/code 200 in 26ms (compile: 17ms, render: 9ms)
app-1    |  GET /api/channels/code 200 in 30ms (compile: 10ms, render: 20ms)
```

**TC-17: Non-admin attempts admin delete**
**Expected:** 403 returned, action blocked by UI
**Steps:** while still logged in as ‘matt’, attempt to delete the post ‘check this out!’ in the channel ‘code’.
**Actual:** UI does not render delete button, no URL manipulation can perform the action.

**TC-18: Sign out**
**Expected:** user routed to landing page, write actions blocked until sign in again
**Steps:** press sign out button while logged in as ‘matt’
**Actual:** logout success, routed back to home page
**Output:**

```
app-1    |  GET /api/auth/csrf 200 in 47ms (compile: 37ms, render: 10ms)
app-1    |  POST /api/auth/signout 200 in 10ms (compile: 3ms, render: 7ms)
app-1    |  GET / 200 in 65ms (compile: 6ms, render: 59ms)
```

**TC-19: Data Persistence**
**Expected:** all test data previously created persists after running ‘docker compose down’,‘docker compose up’ and reloading browser
**Steps:** docker compose down, docker compose up, reload browser, check for user ‘matt’ with sign in, check for channel ‘code’
**Actual:** all previously created test data exists on site, user ‘matt’ exists and can be logged in as, channel ‘code’ and its posts persists
**Output:**

```
matt@Matts-MacBook-Air project % docker compose down
[+] down 3/3
 ✔ Container project-app-1 Removed
0.5s
 ✔ Container postgres      Removed                                    0.2s
 ✔ Network project_default Removed
0.2s
matt@Matts-MacBook-Air project % docker compose up
[+] up 3/3
 ✔ Network project_default Created                                    0.0s
 ✔ Container postgres      Created                                    0.0s
 ✔ Container project-app-1 Created                                    0.0s
Attaching to postgres, app-1
Container postgres Waiting
```

```
postgres  |
postgres  | PostgreSQL Database directory appears to contain a database; Skipping initialization
postgres  |
postgres  | 2026-03-28 15:17:51.836 UTC [1] LOG:  starting PostgreSQL 16.13 (Debian 16.13-1.pgdg13+1) on aarch64-unknown-linux-gnu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
postgres  | 2026-03-28 15:17:51.836 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
postgres  | 2026-03-28 15:17:51.836 UTC [1] LOG:  listening on IPv6 address "::", port 5432
postgres  | 2026-03-28 15:17:51.837 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
postgres  | 2026-03-28 15:17:51.838 UTC [29] LOG:  database system was shut down at 2026-03-28 15:17:48 UTC
postgres  | 2026-03-28 15:17:51.840 UTC [1] LOG:  database system is ready to accept connections
Container postgres Healthy
app-1     |
app-1     | > project@0.1.0 dev
app-1     | > next dev
app-1     |
app-1     | ▲ Next.js 16.1.6 (Turbopack)
app-1     | - Local:        http://localhost:3000
app-1     | - Network:      http://172.18.0.3:3000
app-1     |
app-1     | ✓ Starting...
app-1     | Attention: Next.js now collects completely anonymous telemetry regarding usage.
app-1     | This information is used to shape Next.js' roadmap and prioritize features.
app-1     | You can learn more, including how to opt-out if you'd not like to participate in this anonymous program, by visiting the following URL:
app-1     | https://nextjs.org/telemetry
app-1     |
app-1     | ✓ Ready in 372ms
app-1     |  GET / 200 in 1650ms (compile: 1493ms, render: 157ms)
app-1     |  GET / 200 in 47ms (compile: 7ms, render: 40ms)
app-1     |  GET /login 200 in 311ms (compile: 300ms, render: 10ms)
app-1     |  GET /api/auth/providers 200 in 683ms (compile: 673ms, render: 10ms)
app-1     |  GET /api/auth/csrf 200 in 8ms (compile: 5ms, render: 3ms)
```

(I put in the wrong password here on accident)

```
app-1     | [auth][error] CredentialsSignin: Read more at https://errors.authjs.dev#credentialssignin
app-1     |     at Module.callback (/app/.next/dev/server/chunks/node_modules_@auth_core_36df0127._.js:4314:30)
```

```
app-1  |  at async AuthInternal (/app/.next/dev/server/chunks/node_modules_@auth_core_36df0127._.js:5148:24)
app-1  |  at async Auth (/app/.next/dev/server/chunks/node_modules_@auth_core_36df0127._.js:5370:34)
app-1  |  at async AppRouteRouteModule.do (/app/node_modules/next/dist/compiled/next-server/app-route-turbo.runtime.dev.js:5:37866)
app-1  |  at async AppRouteRouteModule.handle (/app/node_modules/next/dist/compiled/next-server/app-route-turbo.runtime.dev.js:5:45156)
app-1  |  at async responseGenerator (/app/.next/dev/server/chunks/node_modules_next_c0d7f167._.js:17154:38)
app-1  |  at async AppRouteRouteModule.handleResponse (/app/node_modules/next/dist/compiled/next-server/app-route-turbo.runtime.dev.js:1:191938)
app-1  |  at async handleResponse (/app/.next/dev/server/chunks/node_modules_next_c0d7f167._.js:17217:32)
app-1  |  at async Module.handler (/app/.next/dev/server/chunks/node_modules_next_c0d7f167._.js:17270:13)
app-1  |  at async DevServer.renderToResponseWithComponentsImpl (/app/node_modules/next/dist/server/base-server.js:1442:9)
app-1  |  at async DevServer.renderPageComponent (/app/node_modules/next/dist/server/base-server.js:1494:24)
app-1  |  at async DevServer.renderToResponseImpl (/app/node_modules/next/dist/server/base-server.js:1544:32)
app-1  |  at async DevServer.pipeImpl (/app/node_modules/next/dist/server/base-server.js:1038:25)
app-1  |  at async NextNodeServer.handleCatchallRenderRequest (/app/node_modules/next/dist/server/next-server.js:395:17)
app-1  |  at async DevServer.handleRequestImpl (/app/node_modules/next/dist/server/base-server.js:929:17)
app-1  |  at async /app/node_modules/next/dist/server/dev/next-dev-server.js:387:20
app-1  |  at async Span.traceAsyncFn (/app/node_modules/next/dist/trace/trace.js:157:20)
app-1  |  at async DevServer.handleRequest (/app/node_modules/next/dist/server/dev/next-dev-server.js:383:24)
app-1  |  at async invokeRender (/app/node_modules/next/dist/server/lib/router-server.js:248:21)
app-1  |  at async handleRequest (/app/node_modules/next/dist/server/lib/router-server.js:447:24)
app-1  |  at async requestHandlerImpl (/app/node_modules/next/dist/server/lib/router-server.js:496:13)
app-1  |  at async Server.requestListener (/app/node_modules/next/dist/server/lib/start-server.js:226:13)
app-1  | POST /api/auth/callback/credentials? 200 in 213ms (compile: 3ms, render: 210ms)
```

app-1 | GET /api/auth/session 200 in 5ms (compile: 1737µs, render: 4ms)
app-1 | GET /api/auth/providers 200 in 23ms (compile: 16ms, render: 7ms)
app-1 | GET /api/auth/csrf 200 in 5ms (compile: 1740µs, render: 3ms)
app-1 | POST /api/auth/callback/credentials? 200 in 190ms (compile: 1666µs, render: 188ms)
app-1 | GET /api/auth/session 200 in 7ms (compile: 1244µs, render: 5ms)
app-1 | GET /channels 200 in 293ms (compile: 280ms, render: 12ms)
app-1 | GET /channels 200 in 42ms (compile: 14ms, render: 28ms)
app-1 | GET /api/channels 200 in 102ms (compile: 96ms, render: 6ms)
app-1 | GET /api/channels 200 in 110ms (compile: 101ms, render: 9ms)
app-1 | GET /channels/code 200 in 464ms (compile: 454ms, render: 10ms)
app-1 | GET /api/channels/code 200 in 525ms (compile: 519ms, render: 5ms)
app-1 | GET /api/channels/code 200 in 526ms (compile: 520ms, render: 6ms)

**Other Tests**

**Database Seed Behaviour**
Initial Database Seed Verification:
While running:

```
docker exec -it postgres psql -U admin -d proj
```

Query with:

```
proj=# SELECT * FROM users;
```

Result:

```
 id |                    password_hash                    | display_name | role
----+-----------------------------------------------------+--------------+-------
  1 | $2b$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi | admin
| admin
  2 | $2b$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi | alice
| user
  3 | $2b$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi | bob
| user
  4 | $2b$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi | carol
| user
(4 rows)
```

**Password Encryption**
I manually verified the password encryption by creating a new user through the frontend, then running:

```
docker exec -it postgres psql -U admin -d proj
proj=# SELECT display_name, password_hash FROM users;
```

Which gave me:

```
 display_name |                    password_hash
--------------+-----------------------------------------------------
```

```
 admin          |
$2b$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi
 alice          |
$2b$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi
 bob            |
$2b$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi
 carol          |
$2b$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi
 mrpresident    |
$2b$12$ql1rHk9t2z3Y3V.Xk620zuyfPx60Tg3lYbc6B/XBSppelswZr0ZWS
```

The first three are seeded users, the last one is my new account. As you can see, the seeded users used ten rounds of bcrypt decryption, while the new one used twelve.